

# ATF Field Specification v1.3

OMNIX QUANTUM LTD · Harold Alberto Nunes Rodelo · Day 1 Deliverable for VeriSigil AI · 22 May 2026

ATF FIELD SPECIFICATION Agent Trust Fabric — Trace Format & Field Definitions Version: 1.3 ·  
Date: 2026-05-22 Issued by: Harold Alberto Nunes Rodelo, ATF / OMNIX QUANTUM LTD For:  
VeriSigil AI (Raheem Larry Babatunde) — Bridge Validation Collaboration Status: Day 1  
Deliverable — Sandbox / Evaluation Only

## Overview

This document defines the complete field specification for the three core ATF record types used in the TAR↔TAP and RCR↔Survivability bridge validation:

- DR — Delegation Receipt (ATFDR-`{16HEX}`)
- TAR — Temporal Admissibility Record (ATFTAR-`{16HEX}`)
- RCR — Runtime Continuity Record (ATFRCR-`{16HEX}`)

All records are PQC-signed (Dilithium-3 / ML-DSA-65, FIPS 204). All `content_hash` values are SHA-256 hex digests (no prefix). All timestamps are ISO 8601 UTC unless noted as nanosecond epoch.

## 1. Delegation Receipt (DR)

Record ID format: `ATFDR-{16 uppercase hex characters}` ADR reference: ADR-156

### 1.1 Fields

| Field                                   | Type   | Required | Description                                                                                                       |
|-----------------------------------------|--------|----------|-------------------------------------------------------------------------------------------------------------------|
| <code>delegation_id</code>              | string | REQUIRED | Unique DR identifier. Format: <code>ATFDR-<code>{16HEX}</code></code>                                             |
| <code>delegator_id</code>               | string | REQUIRED | AID or human operator ID issuing the delegation                                                                   |
| <code>delegate_id</code>                | string | REQUIRED | AID of the receiving (delegated) agent                                                                            |
| <code>task_scope</code>                 | object | REQUIRED | Dict describing what the delegate is authorized to do. Arbitrary key-value pairs.                                 |
| <code>authority_budget_delegator</code> | float  | REQUIRED | Authority budget held by delegator at issuance. Range: 0.0-100.0                                                  |
| <code>authority_budget_granted</code>   | float  | REQUIRED | Authority budget granted to delegate. MUST be $\leq$ <code>authority_budget_delegator</code> (ATF-INV-001 / MAR). |
| <code>parent_delegation_id</code>       | string | OPTIONAL | DR ID of the delegation that gave delegator its authority. Null if delegator is human root.                       |

| Field                | Type    | Required    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------|---------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| chain_root_id        | string  | REQUIRED    | DR ID of the originating human-root delegation in the chain.                                                                                                                                                                                                                                                                                                                                                                                                |
| delegation_depth     | integer | REQUIRED    | Chain depth. 0 = human root, 1 = first agent, N = Nth sub-agent.                                                                                                                                                                                                                                                                                                                                                                                            |
| delegator_public_key | string  | REQUIRED    | Dilithium-3 public key of delegator, base64-encoded. Used for offline DR signature verification. May be empty string "" when signing key was not available at issuance time — treat empty string as unsigned.                                                                                                                                                                                                                                               |
| content_hash         | string  | REQUIRED    | SHA-256 hex digest of all DR fields except content_hash, pqc_signature, pqc_algorithm. Canonical JSON: json.dumps(fields, sort_keys=True, separators=(",", ":"), default=str). Note: authority_budget_delegator and authority_budget_granted are rounded to 4dp before hashing.                                                                                                                                                                             |
| posture_state_hash   | string  | CONDITIONAL | SHA-256 of authority posture at exact moment of DR issuance. Present in all DRs issued after ATF v1.0. Null only in legacy pre-v1.0 records. Committed fields: delegator_id, task_scope, authority_budget_delegator (rounded to 4dp), authority_budget_granted (rounded to 4dp), delegation_depth, chain_root_id, created_at. Canonical JSON: json.dumps(fields, sort_keys=True, separators=(",", ":")). Enables SPV binding for TAR↔TAP bridge validation. |
| pqc_signature        | string  | CONDITIONAL | Dilithium-3 (ML-DSA-65) signature over content_hash, base64-encoded. Present when signing key is available.                                                                                                                                                                                                                                                                                                                                                 |
| pqc_algorithm        | string  | CONDITIONAL | Signing algorithm identifier. Value: "dilithium3"                                                                                                                                                                                                                                                                                                                                                                                                           |
| expires_at           | string  | OPTIONAL    | ISO 8601 UTC expiry timestamp. Null = no expiry.                                                                                                                                                                                                                                                                                                                                                                                                            |
| status               | string  | REQUIRED    | ACTIVE   EXPIRED   REVOKED                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| created_at           | string  | REQUIRED    | ISO 8601 UTC timestamp of DR issuance.                                                                                                                                                                                                                                                                                                                                                                                                                      |
| metadata             | object  | REQUIRED    | Extension dict. Empty object {} if unused.                                                                                                                                                                                                                                                                                                                                                                                                                  |

## 1.2 Invariants

- ATF-INV-001 (MAR): `authority_budget_granted ≤ authority_budget_delegator`. Violation raises hard error before issuance — no DR is created.

- ATF-INV-002: `chain_root_id` traces back to a human-root delegation ( `delegation_depth = 0` ).
- ATF-INV-003: `posture_state_hash` is computed before `content_hash` . `content_hash` commits to `posture_state_hash` .

### 1.3 Example

```
{
  "delegation_id": "ATFDR-A1B2C3D4E5F60001",
  "delegator_id": "AID-COMPLIANCE-0001",
  "delegate_id": "AID-AUDIT-0007",
  "task_scope": {
    "domain": "compliance",
    "permitted_actions": ["read_evidence", "issue_tar"],
    "max_duration_s": 3600
  },
  "authority_budget_delegator": 80.0,
  "authority_budget_granted": 60.0,
  "parent_delegation_id": "ATFDR-R00T0000000001",
  "chain_root_id": "ATFDR-R00T0000000001",
  "delegation_depth": 1,
  "delegator_public_key": "<base64-dilithium3-pubkey>",
  "content_hash": "a3f1c2d4e5b6a7f8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2",
  "posture_state_hash": "2620ba4e69a108faeb74144b906d88c45c489d6c920de888fe0fa333a91eb393",
  "pqc_signature": "<base64-dilithium3-signature>",
  "pqc_algorithm": "dilithium3",
  "expires_at": "2026-05-22T12:00:00+00:00",
  "status": "ACTIVE",
  "created_at": "2026-05-21T10:00:00+00:00",
  "metadata": {}
}
```

## 2. Temporal Admissibility Record (TAR)

Record ID format: `ATFTAR-{16 uppercase hex characters}` ADR reference: ADR-157

### 2.1 Fields

| Field                      | Type    | Required | Description                                                        |
|----------------------------|---------|----------|--------------------------------------------------------------------|
| <code>tar_id</code>        | string  | REQUIRED | Unique TAR identifier. Format: <code>ATFTAR-{16HEX}</code>         |
| <code>delegation_id</code> | string  | REQUIRED | <code>ATFDR-{16HEX}</code> of the authorizing Delegation Receipt   |
| <code>agent_id</code>      | string  | REQUIRED | AID of the acting agent. Format: <code>AID-{DOMAIN}-{16HEX}</code> |
| <code>execution_ref</code> | string  | OPTIONAL | Reference to Execution Receipt (ADR-131). Null if not linked.      |
| <code>execution_ns</code>  | integer | REQUIRED | Nanosecond Unix timestamp of the execution event being admitted.   |
| <code>execution_ts</code>  | string  | REQUIRED | ISO 8601 UTC timestamp of the execution event.                     |

| Field                               | Type   | Required    | Description                                                                                                                       |
|-------------------------------------|--------|-------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <code>dr_status_at_admission</code> | string | REQUIRED    | Status of the DR at the moment of TAR admission. Value: <code>ACTIVE</code>   <code>EXPIRED</code>   <code>REVOKED</code>         |
| <code>dr_expires_at</code>          | string | OPTIONAL    | DR expiry timestamp, copied from the DR.                                                                                          |
| <code>authority_budget</code>       | float  | REQUIRED    | Authority budget of the DR at admission time.                                                                                     |
| <code>domain</code>                 | string | REQUIRED    | Governance domain identifier (e.g. <code>"compliance"</code> , <code>"trading"</code> , <code>"medical_ai"</code> ).              |
| <code>task_action</code>            | string | REQUIRED    | The specific action being admitted (e.g. <code>"read_evidence"</code> , <code>"issue_receipt"</code> ).                           |
| <code>admission_status</code>       | string | REQUIRED    | <code>ADMITTED</code>   <code>REJECTED</code>                                                                                     |
| <code>rejection_reason</code>       | string | OPTIONAL    | Human-readable reason for rejection. Null if admitted.                                                                            |
| <code>content_hash</code>           | string | REQUIRED    | SHA-256 hex digest of all TAR fields except <code>content_hash</code> , <code>pgc_signature</code> , <code>pgc_algorithm</code> . |
| <code>pgc_signature</code>          | string | CONDITIONAL | Dilithium-3 (ML-DSA-65) signature over <code>content_hash</code> , base64-encoded.                                                |
| <code>pgc_algorithm</code>          | string | CONDITIONAL | <code>"dilithium3"</code>                                                                                                         |
| <code>chain_root_id</code>          | string | REQUIRED    | <code>chain_root_id</code> of the authorizing DR. Enables chain-level traceability.                                               |
| <code>issued_at</code>              | string | REQUIRED    | ISO 8601 UTC timestamp of TAR issuance.                                                                                           |
| <code>metadata</code>               | object | REQUIRED    | Extension dict.                                                                                                                   |

## 2.2 Invariants

- ATF-INV-004: A TAR with `admission_status = ADMITTED` MUST reference a DR with `dr_status_at_admission = ACTIVE`.
- ATF-INV-005: `execution_ns` MUST be  $\leq$  the nanosecond timestamp of `issued_at`.
- TAR is the primary bridge target for TAR $\leftrightarrow$ TAP equivalence mapping. `delegation_id` is the join key.

## 2.3 Example

```
{
  "tar_id": "ATFTAR-F1E2D3C4B5A60001",
  "delegation_id": "ATFDR-A1B2C3D4E5F60001",
  "agent_id": "AID-COMPLIANCE-F1E2D3C4B5A60001",
  "execution_ref": null,
  "execution_ns": 1748131200000000000,
  "execution_ts": "2026-05-21T10:00:00+00:00",
  "dr_status_at_admission": "ACTIVE",
  "dr_expires_at": "2026-05-22T12:00:00+00:00",
  "authority_budget": 60.0,
  "domain": "compliance",
  "task_action": "read_evidence",
  "admission_status": "ADMITTED",
```

```
"rejection_reason": null,
"content_hash": "c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6",
"pqc_signature": "<base64-dilithium3-signature>",
"pqc_algorithm": "dilithium3",
"chain_root_id": "ATFDR-R00T0000000001",
"issued_at": "2026-05-21T10:00:01+00:00",
"metadata": {}
}
```

### 3. Runtime Continuity Record (RCR)

Record ID format: `ATFRCR-{16 uppercase hex characters}` ADR reference: ADR-159

#### 3.1 Fields

| Field                          | Type    | Required | Description                                                                                                                                                                                           |
|--------------------------------|---------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>rcr_id</code>            | string  | REQUIRED | Unique RCR identifier. Format: <code>ATFRCR-{16HEX}</code>                                                                                                                                            |
| <code>tar_id</code>            | string  | REQUIRED | <code>ATFTAR-{16HEX}</code> of the admission record this RCR monitors.                                                                                                                                |
| <code>delegation_id</code>     | string  | REQUIRED | <code>ATFDR-{16HEX}</code> of the authorizing DR.                                                                                                                                                     |
| <code>chain_root_id</code>     | string  | REQUIRED | Chain root delegation ID.                                                                                                                                                                             |
| <code>agent_id</code>          | string  | REQUIRED | AID of the running agent.                                                                                                                                                                             |
| <code>ces_score</code>         | float   | REQUIRED | Continuity Eligibility Score. Range: 0.0–100.0. Formula: $(T \times 0.30) + (B \times 0.30) + (D \times 0.20) + (I \times 0.20)$                                                                      |
| <code>continuity_status</code> | string  | REQUIRED | <code>NOMINAL</code> (75–100)   <code>MONITORING</code> (50–75)   <code>WARNING</code> (25–50)   <code>CRITICAL</code> (10–25)   <code>HALT</code> (0–10)                                             |
| <code>ces_temporal</code>      | float   | REQUIRED | T component: $(\text{time remaining on DR} / \text{total DR lifetime}) \times 100$ . Range: 0.0–100.0. Defaults to 100.0 when DR has no expiry ( <code>dr_expires_at</code> is null). Rounded to 4dp. |
| <code>ces_budget</code>        | float   | REQUIRED | B component: $(\text{budget remaining} / \text{budget at admission}) \times 100$ . Range: 0.0–100.0. Defaults to 100.0 when <code>budget_at_admission = 0</code> . Rounded to 4dp.                    |
| <code>ces_context</code>       | float   | REQUIRED | D component: $100 - \text{context\_drift\_pct}$ . Range: 0.0–100.0. Rounded to 4dp.                                                                                                                   |
| <code>ces_integrity</code>     | float   | REQUIRED | I component: $100 - (\text{active\_anomalies} \times 10)$ . Range: 0.0–100.0. Rounded to 4dp.                                                                                                         |
| <code>time_remaining_ns</code> | integer | OPTIONAL | Remaining DR lifetime in nanoseconds at this snapshot. Null when DR has no expiry ( <code>dr_expires_at</code> is null).                                                                              |
| <code>dr_expires_at</code>     | string  | OPTIONAL | ISO 8601 UTC expiry timestamp of the DR. Null if DR has no expiry.                                                                                                                                    |

| Field               | Type    | Required    | Description                                                                                                                                                                      |
|---------------------|---------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fragmentation_score | float   | REQUIRED    | Aggregate budget consumption % across all active sessions sharing the same chain_root_id . Formula: (total_admitted - total_remaining) / total_admitted × 100 . Range: 0.0–100.0 |
| reauth_challenge_id | string  | OPTIONAL    | Reference to ReauthorizationChallenge if issued ( ATFCRC- {16HEX} ). Null otherwise.                                                                                             |
| execution_ns        | integer | REQUIRED    | Nanosecond Unix timestamp of the execution event.                                                                                                                                |
| execution_ts        | string  | REQUIRED    | ISO 8601 UTC timestamp of the execution event.                                                                                                                                   |
| issued_at           | string  | REQUIRED    | ISO 8601 UTC timestamp of RCR issuance.                                                                                                                                          |
| budget_at_admission | float   | REQUIRED    | Authority budget at the moment of TAR admission.                                                                                                                                 |
| budget_remaining    | float   | REQUIRED    | Authority budget remaining at this RCR snapshot.                                                                                                                                 |
| context_drift_pct   | float   | REQUIRED    | Context drift percentage at this snapshot. Range: 0.0–100.0                                                                                                                      |
| active_anomalies    | integer | REQUIRED    | Count of active governance anomalies at this snapshot.                                                                                                                           |
| content_hash        | string  | REQUIRED    | SHA-256 hex digest of all RCR fields except content_hash , pqc_signature , pqc_algorithm .                                                                                       |
| pqc_signature       | string  | CONDITIONAL | Dilithium-3 (ML-DSA-65) signature over content_hash , base64-encoded.                                                                                                            |
| pqc_algorithm       | string  | CONDITIONAL | "dilithium3"                                                                                                                                                                     |
| sample_reason       | string  | OPTIONAL    | Reason for this RCR snapshot: SCHEDULED   EXECUTION_COMPLETE   THRESHOLD_BREACH   FRAGMENTATION   EXTERNAL   RC_TTL_EXPIRED .                                                    |
| escalation_event_id | string  | OPTIONAL    | ID of ContinuityEscalationEvent if triggered. Format: ATFCEE- {16HEX} . Null otherwise.                                                                                          |
| predecessor_rcr_id  | string  | OPTIONAL    | ID of previous RCR in the continuity chain.                                                                                                                                      |
| metadata            | object  | REQUIRED    | Extension dict.                                                                                                                                                                  |

### 3.2 Invariants

- RGC-INV-001: Every RCR MUST be anchored to a valid TAR ( tar\_id must resolve).
- RGC-INV-002: CES MUST be computed from real-time values only — no cached inputs.
- RGC-INV-003: HALT tier terminates execution and revokes in-flight sub-tasks.
- RGC-INV-005: All RCRs are PQC-signed and immutable after issuance.
- RGC-INV-006: RCR chain is acyclic — issued\_at MUST be strictly monotonically increasing per execution session.
- RCR is the bridge target for RCR↔Survivability receipt mapping. Join key: tar\_id .

### 3.3 Example

```
{
  "rcr_id": "ATFRCR-D1C2B3A4E5F60001",
  "tar_id": "ATFTAR-F1E2D3C4B5A60001",
  "delegation_id": "ATFDR-A1B2C3D4E5F60001",
  "chain_root_id": "ATFDR-ROOT0000000001",
  "agent_id": "AID-COMPLIANCE-F1E2D3C4B5A60001",
  "ces_score": 92.7,
  "continuity_status": "NOMINAL",
  "ces_temporal": 87.0,
  "ces_budget": 92.0,
  "ces_context": 95.0,
  "ces_integrity": 100.0,
  "execution_ns": 1748131200000000000,
  "execution_ts": "2026-05-21T10:00:00+00:00",
  "issued_at": "2026-05-21T10:01:00+00:00",
  "budget_at_admission": 60.0,
  "budget_remaining": 55.2,
  "context_drift_pct": 5.0,
  "active_anomalies": 0,
  "time_remaining_ns": 936000000000000,
  "dr_expires_at": "2026-05-22T12:00:00+00:00",
  "fragmentation_score": 8.0,
  "reauth_challenge_id": null,
  "content_hash": "d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6e7",
  "pqc_signature": "<base64-dilithium3-signature>",
  "pqc_algorithm": "dilithium3",
  "sample_reason": "SCHEDULED",
  "escalation_event_id": null,
  "predecessor_rcr_id": null,
  "metadata": {}
}
```

## 4. Cross-Record Traceability

The three records form a traceable chain:

```
DR (ATFDR-...)
├─ TAR (ATFTAR-...) – references delegation_id
│   └─ RCR (ATFRCR-...) – references tar_id + delegation_id
```

Join path for TAR↔TAP bridge: TAR.delegation\_id → DR.delegation\_id →  
DR.posture\_state\_hash

Join path for RCR↔Survivability bridge: RCR.tar\_id → TAR.tar\_id → TAR.delegation\_id →  
DR.chain\_root\_id

## 5. Signature Verification

Steps 1-3 are identical for all three record types:

1. Recompute `content_hash` from all fields except `content_hash` , `pqc_signature` , `pqc_algorithm`
2. Canonical JSON: `json.dumps(fields, sort_keys=True, separators=(",", ":"), default=str)`
3. SHA-256 hex of UTF-8 encoded canonical JSON

Step 4 differs by record type:

**DR:** Verify Dilithium-3 signature over `content_hash.encode("utf-8")` using `DR.delegator_public_key` (the key is embedded in the DR itself — no external key required).

**TAR and RCR:** Verify Dilithium-3 signature over `content_hash.encode("utf-8")` using the OMNIX platform public key.

OMNIX platform public key (ML-DSA-65 / Dilithium-3, base64): Shared separately. See integration comms for current value.

## 6. Domain Values (Common)

Recognized `domain` values:

| Value                           | Description                          |
|---------------------------------|--------------------------------------|
| <code>compliance</code>         | ISO 27001 / regulatory compliance    |
| <code>trading</code>            | Autonomous trading governance        |
| <code>medical_ai</code>         | Medical AI recommendation governance |
| <code>defense_governance</code> | Defense / national security AI       |
| <code>insurance</code>          | Insurance underwriting AI            |
| <code>real_estate</code>        | Real estate AI decisions             |
| <code>energy_governance</code>  | Energy sector AI                     |
| <code>islamic_credit</code>     | Islamic finance credit AI            |
| <code>stablecoin</code>         | Stablecoin / DeFi governance         |
| <code>crisis</code>             | Crisis response AI                   |
| <code>autonomous_agent</code>   | General autonomous agent             |
| <code>robotics</code>           | Robotics governance                  |

## 7. Revision History

| Version | Date       | Change                                                              |
|---------|------------|---------------------------------------------------------------------|
| 1.0     | 2026-05-21 | Initial release — Day 1 deliverable for VeriSigil bridge validation |
| 1.1     | 2026-05-22 |                                                                     |



| Version | Date       | Change                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |            | RCR field names corrected to match implementation: ces→ces_score, ces_tier→continuity_status, temporal_health→ces_temporal, budget_health→ces_budget, context_fidelity→ces_context, integrity_score→ces_integrity. Added budget_at_admission, budget_remaining, context_drift_pct, active_anomalies, sample_reason, escalation_event_id, predecessor_rcr_id.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1.2     | 2026-05-22 | CES component ranges corrected: 0.0-1.0 → 0.0-100.0 for ces_temporal, ces_budget, ces_context, ces_integrity. Component formulas corrected (no / 100 division). Added missing RCR fields: time_remaining_ns, dr_expires_at, fragmentation_score, reauth_challenge_id. Example JSON values corrected. Fixed RGC-INV-006 reference from issued_at_ns to issued_at. DR signature verification corrected: uses DR.delegator_public_key, not platform key. TAR/RCR use platform key. Canonical JSON updated to include default=str for all three record types.                                                                                                                                                                                                                                                                                                                |
| 1.3     | 2026-05-22 | sample_reason values corrected: removed ANOMALY_DETECTED/MANUAL, added THRESHOLD_BREACH/FRAGMENTATION/EXTERNAL/RC_TTL_EXPIRED. fragmentation_score range corrected: 0.0-1.0 → 0.0-100.0, formula documented. Example fragmentation_score corrected: 0.08 → 8.0. posture_state_hash: added float rounding (4dp), canonical JSON format, status changed to CONDITIONAL. ces_score example corrected: 82.5 → 92.7. posture_state_hash example hash replaced with real computed value. DR signature uses delegator_public_key (not platform key); added empty-string note. CES components: documented 4dp rounding and ces_temporal=100.0 default when no expiry. time_remaining_ns corrected to OPTIONAL. escalation_event_id format ATFCEE-{16HEX} added. robotics domain added. canonical JSON updated to include default=str for content_hash in all three record types. |